department	total_patients
Geriatrics	20
Cardiology	19
Oncology	17
Orthopedics	16
Pediatrics	13
Dermatology	12
Surgery	12
Neurology	9

Q1. Which department has most patients?

Geriatrics has most patients (20), followed by Cardiology (19). Neurology has the least (9).

Hospital handles more old-age patients.

department	avg_stay
Surgery	11
Oncology	9
Geriatrics	8
Cardiology	6
Orthopedics	5
Neurology	4
Pediatrics	3
Dermatology	2

Q2. Which department has longest vs shortest average stay?

Surgery stays longest (10.6 days), Oncology second (9.3 days). Dermatology stays shortest (2.4 days).

Surgical cases need more recovery time.

doctor_name	total_cases	avg_stay
Dr. Villanueva	20	8
Dr. Garcia	17	9
Dr. Tan	16	5
Dr. Lim	13	3
Dr. Ramos	12	2
Dr. Lopez	12	11
Dr. Dela Cruz	11	6
Dr. Chua	9	4
Dr. Santos	8	7

Q3. Which doctor handles most cases?

Dr. Villanueva handles most cases (20 patients), Dr. Garcia next (17).

All doctors under 20 cases, workload is fairly distributed but Geriatrics doctor is busiest.

patient_age	avg_stay
21	2
22	1
25	2
26	2
27	3
28	3
29	2
30	4
31	3
32	3
33	3
34	3
35	4
36	4
37	3
38	4
39	4
40	5
41	5
42	9
43	9
44	6
45	5
46	7
47	12
48	5
49	6
50	6
52	9
53	7
54	5
55	5
56	7
57	6
58	8
59	8
60	8
61	7
62	9
63	10
64	9
65	9
66	10
67	9
68	7
70	8
71	9
72	9
73	9
74	10

Q4. Does patient age affect stay?

Yes. Young patients (age 24-29) stay only 1-2 days. Stay gets longer as age increases.

Result shows age 24 = 2 days vs older ages longer.

department	total_revenue (PHP)
Oncology	430800
Cardiology	410300
Surgery	350200
Orthopedics	284900
Geriatrics	222900
Neurology	128000
Pediatrics	124800
Dermatology	96700

Q5. Which department brings most revenue?

Oncology brings most revenue (430,800), Cardiology second (410,300). Dermatology lowest (96,700).

Cancer and heart treatments are most expensive.

department	total_treatment (PHP)	total_medicine (PHP)
Cardiology	341700	68600
Geriatrics	181000	41900
Orthopedics	239400	45500
Pediatrics	103200	21600
Dermatology	82500	14200
Neurology	105600	22400
Oncology	350200	80600
Surgery	288000	62200

Q6. For each department, how much is treatment vs medicine?

Treatment cost is always much higher than medicine cost. Example: Cardiology treatment 341,700 vs medicine only 68,600.

Main income is from treatment rather than

patient_id	length_of_stay	total_cost (PHP)	cost_per_day (PHP)
116	1	8200	8200
145	2	9900	4950
110	2	9800	4900
181	2	9600	4800
130	2	8700	4350
102	5	21000	4200
159	2	8300	4150
168	2	8100	4050
174	2	7900	3950
202	2	7800	3900
146	4	15200	3800
141	5	19000	3800
213	6	22800	3800
122	6	22700	3783
184	5	18900	3780
105	5	18500	3700
160	5	18500	3700
201	6	22100	3683
109	4	14500	3625
192	5	18000	3600
107	2	7200	3600
180	6	21600	3600
175	5	18000	3600
119	4	14400	3600
206	5	18000	3600
101	6	21500	3583
161	6	21500	3583
215	6	21500	3583
216	4	14300	3575
218	5	17800	3560
140	6	21300	3550
190	4	14200	3550
138	5	17700	3540
151	6	20900	3483
166	5	17200	3440
106	3	10300	3433
132	5	17100	3420
186	6	20300	3383
125	5	16900	3380
212	5	16900	3380
167	4	13500	3375
136	3	10100	3366
150	5	16800	3360
189	3	10000	3333
199	5	16600	3320
169	6	19600	3266
171	9	29200	3244
113	6	19300	3216
176	7	22500	3214
187	7	22400	3200

Q7. Which is more profitable per day - short or long stay?

Short stay 1-2 days is more profitable per day. Short stay range is 3,600 to 8,200 per day (avg 4,600), while long stay 4-6 days range is 3,600 to 3,800 per day (avg 3,700).

Short stays = higher daily rate.

patient_age	avg_stay	avg_cost (PHP)
47	12	30700
52	9	26900
66	10	26600
63	10	25300
43	9	25033
60	8	24433
65	9	24425
64	9	24233
46	7	24050
59	8	23675
42	9	23400
58	8	23033
56	7	21900
49	6	21133
53	7	21100
57	6	20300
62	9	20267
44	6	20200
61	7	19500
45	5	19133
67	9	19000
50	6	18600
55	5	17700
54	5	17550
40	5	15800
48	5	15550
41	5	15450
38	4	14733
36	4	14700
30	4	14400
35	4	12367
72	9	11350
73	9	11300
71	9	11100
70	8	11067
68	7	11000
74	10	10900
75	9	10900
37	3	10300
39	4	10300
31	3	10000
32	3	9650
26	2	9600
33	3	9400
29	2	9267
27	3	8900
28	3	8750
21	2	8700
22	1	8200
34	3	7800

8. Are older patients more expensive?

Partially. Older patients cost more until 65y, but after 70y cost drops dramatically.

Most expensive is 47y (₱30,700, 12 days). Top 10 are all 43-65y.